# IAS PRELIMS/MAINS

# SCIENCE & TECHNOLOGY

## A COMPREHENSIVE COVERAGE

## MODULE - 14

## BIOTECHNOLOGY - 4

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# BIOTECHNOLOGY

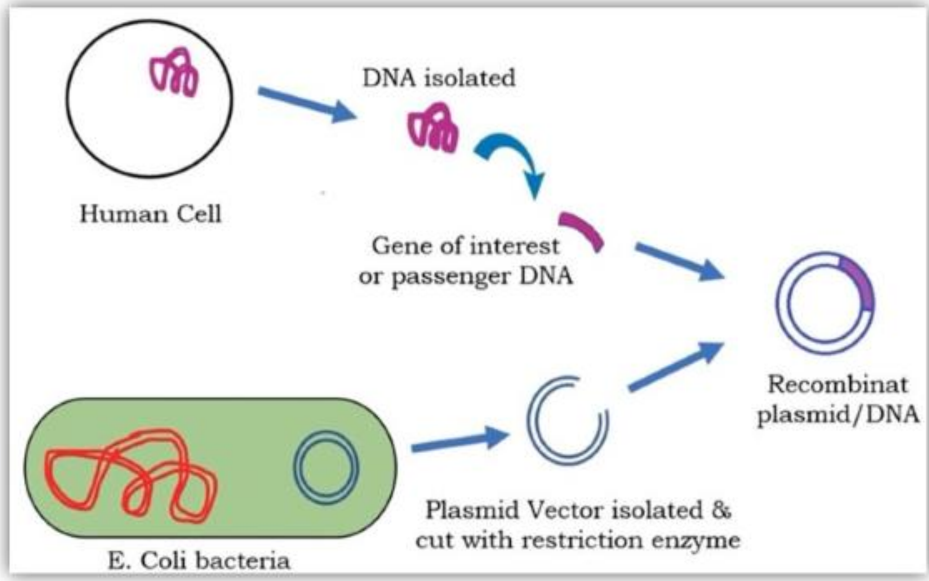
## RECOMBINANT DNA TECHNOLOGY

### DEFINITION 1

Recombinant DNA (rDNA) molecules are DNA molecules formed by laboratory methods of genetic recombination to bring together genetic material from multiple sources and creating sequences that would not otherwise be found in the genome.

### DEFINITION 2

rDNA technology is joining together of DNA molecules from two different species, which is then inserted into a host organism to produce genetic combinations that are of value to science, medicine, agriculture and industry.



# BIOTECHNOLOGY

## rDNA TECHNOLOGY

1. rDNA technology is joining together of DNA Recombinant DNA is possible because DNA molecules from all organisms share the same chemical structure. They only differ in the nucleotide sequence within the overall identical structure.

2. rDNA technology requires the use of molecular scissors called restriction enzymes which cut DNA at specific sequences..

3. The cut part is then inserted into a piece of bacterial DNA called Plasmid. The plasmid is then reintroduced into a bacterial cell.

4. So, rDNA involves the insertion of gene of interest into a vector such as a Plasmid to form a recombinant DNA molecule for the production of large quantities of that gene fragment or product encoded by that gene from two different species, which is then inserted into a host organism to produce genetic combinations that are of value to science, medicine, agriculture and industry.

# SCIENCE AND TECHNOLOGY

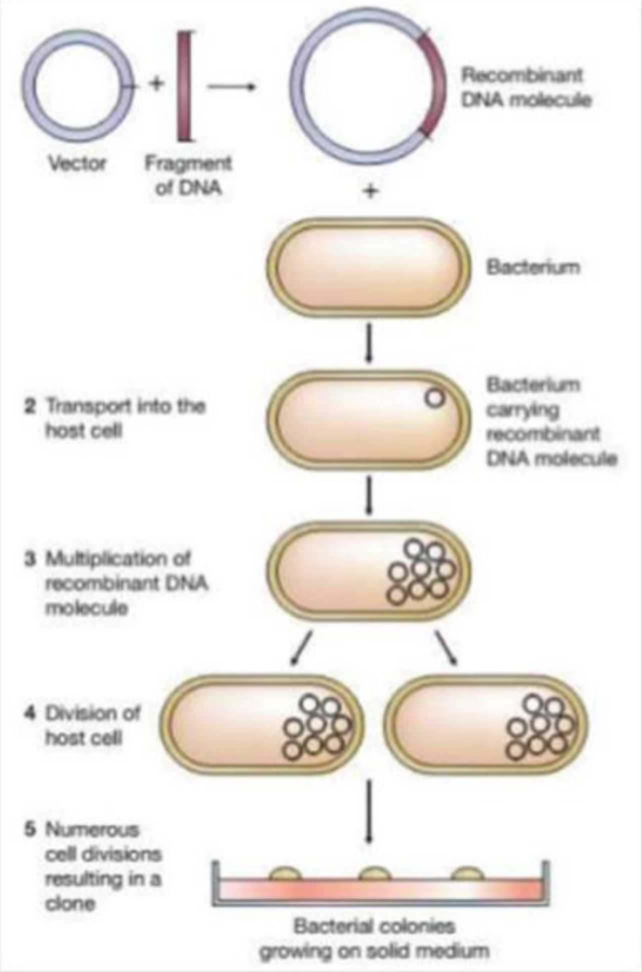
## BIOTECHNOLOGY

## STEPS IN rDNA

STEP 1

### IDENTIFICATION & ISOLATION OF GENE OF INTEREST

- From where do we get the desired gene? We get the desired gene from a Genome Library or we can go for Chemical Synthesis of gene if we know the nucleotide sequence.
- The cutting of DNA at specific locations became possible with the discovery of restriction enzymes which act as molecular scissors.
- Restriction enzymes cut DNA at specific sequences.
- If DNA from different sources is cut with the same restriction enzyme, the cut ends can be joined together to form a continuous DNA piece by enzyme ligase.



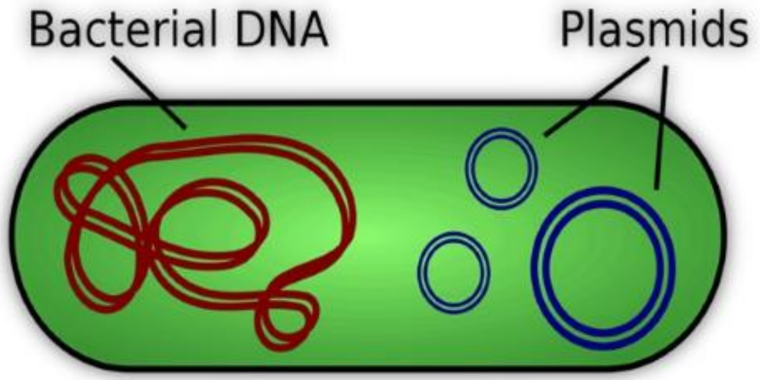
# BIOTECHNOLOGY

## STEPS IN rDNA

STEP 2

### JOINING OF THE GENE INTO A SUITABLE VECTOR (FORMING RECOMBINANT DNA)

- A vector is any DNA molecule that is capable of multiplying inside the host to which our gene of interest is attached. So, a vector DNA is used to deliver a foreign piece of DNA into the host organism.
- Commonly used vectors are plasmids which are circular bacterial DNA.
- Plasmids are cut using restriction enzymes and ligase enzyme is used for joining the vector DNA with the gene of interest.
- The resulting DNA is called the 'recombinant DNA' or 'Chimera' or 'recombinant vector'.



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## BIOTECHNOLOGY

## STEPS IN rDNA

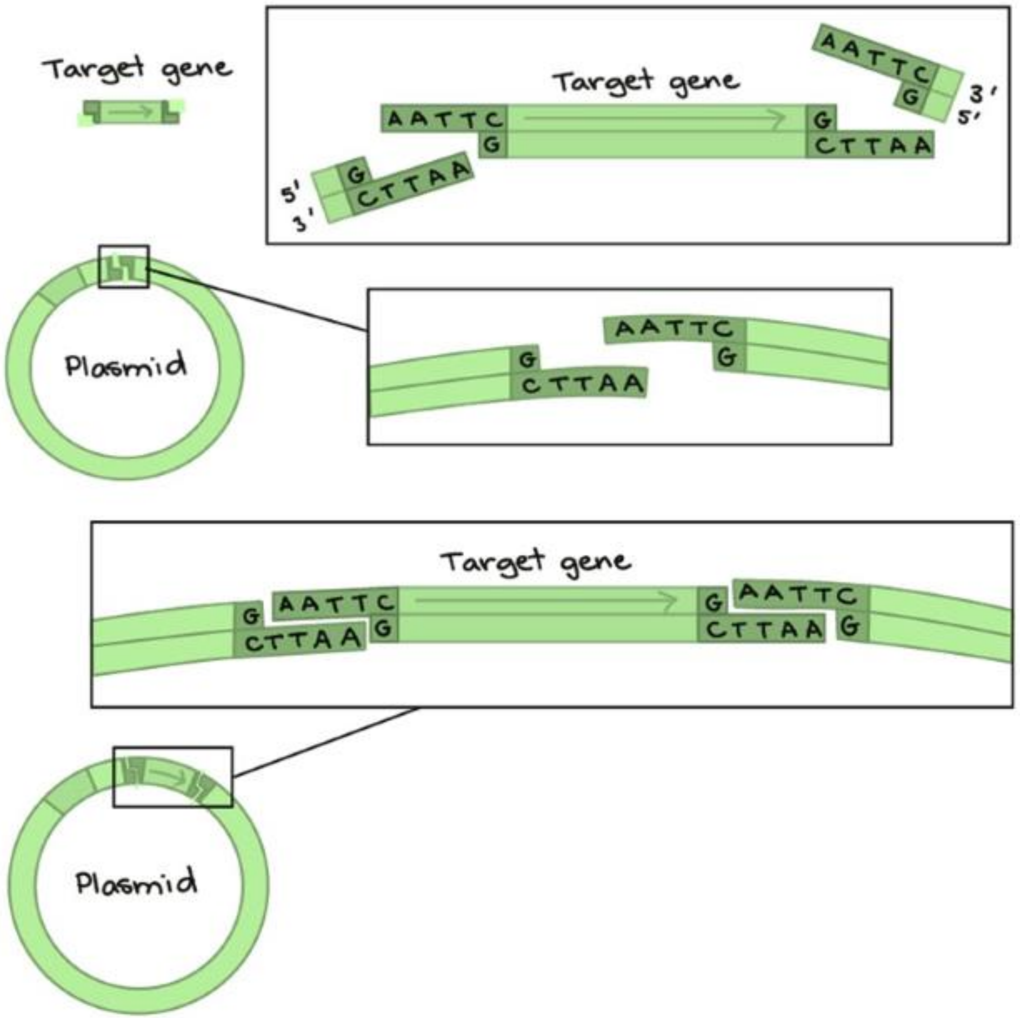
STEP

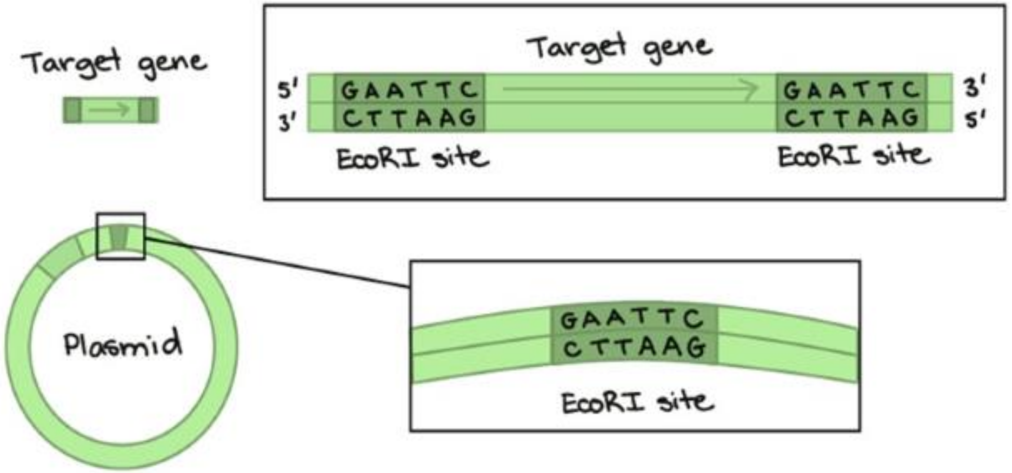
### 3 INTRODUCTION OF THE VECTOR INTO THE HOST ORGANISM

### 4 MULTIPLICATION OF GENE OF INTEREST

- It is also called Gene Cloning.
- The objective of Gene Cloning is either to make numerous copies of the desired gene or to produce the protein encoded by the desired gene.
- The inserted gene along with the vector will replicate inside the host so that many copies of the desired gene are produced.

# STEPS IN rDNA



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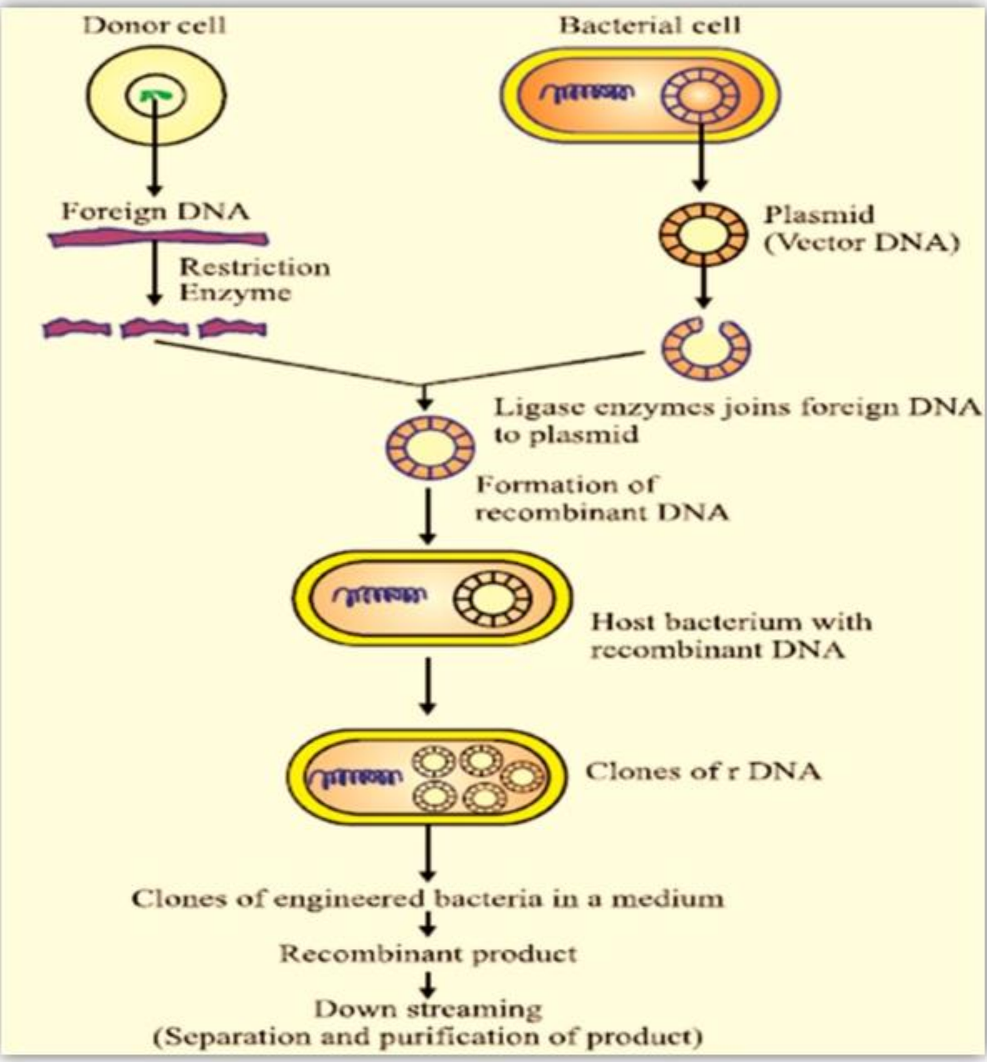
## TOOLS OF THE RDNA TECHNOLOGY

1. **Restriction enzymes** - These are used to cut DNA molecules at specific sequences into many smaller DNA fragments.
2. **Plasmids** - These are circular DNA present in the bacteria, which can replicate independently. Foreign DNA can be placed into a plasmid and it is replicated further.
3. **DNA ligase** - This enzyme is used to join the two pieces of DNA together.
4. **Foreign DNA** - This is also known as passenger DNA, which contains desired gene sequences.
5. **Vector** - It is a vehicle used to insert the desired DNA into the host cell. Plasmid DNA is one of the most widely used vectors.



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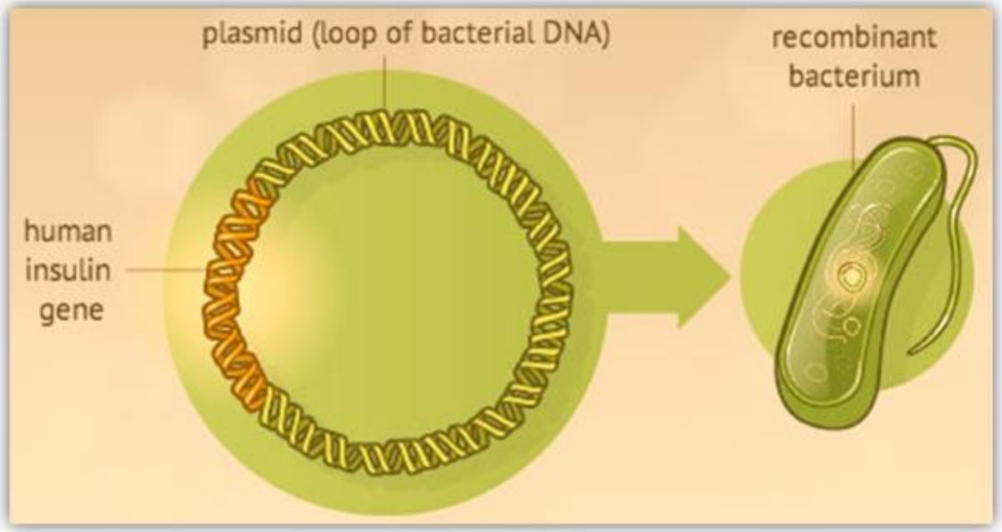
## APPLICATIONS OF rDNA

rDNA provided a way for scientists to produce insulin in labs.

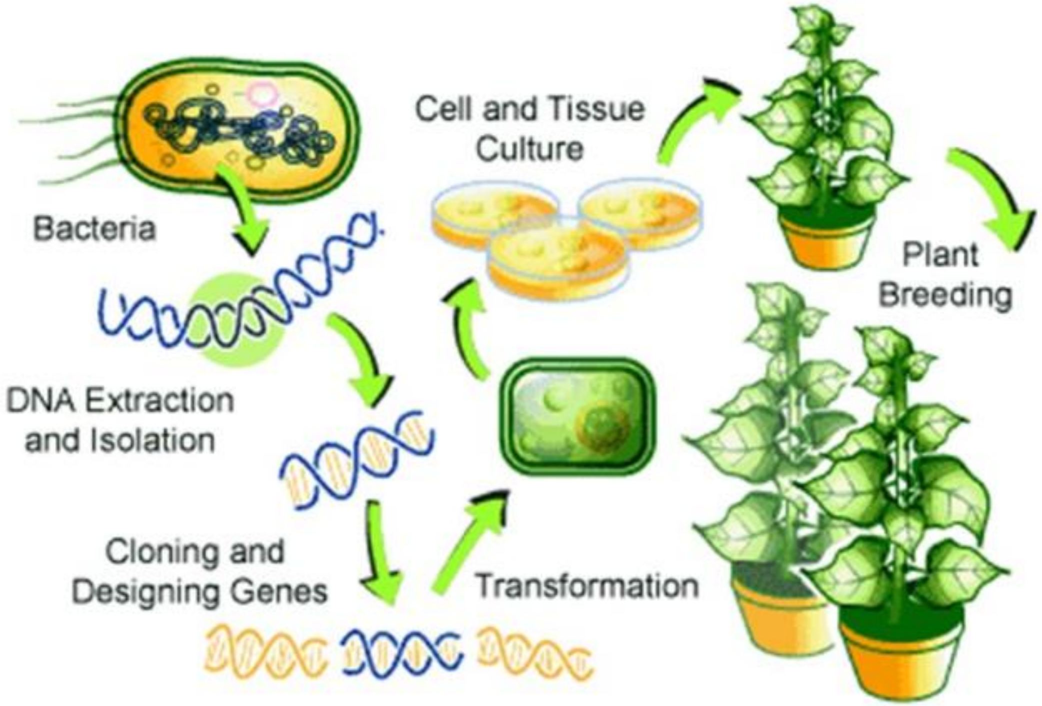
- ✓ The gene for human insulin is isolated from human cells and inserted into plasmid. The plasmids are then introduced into bacterial cells which then manufacture insulin based on the human genetic code.
- ✓ Such insulin is identical to natural human insulin.
- ✓ Scientists also use yeast cells to produce insulin.



## APPLICATIONS OF rDNA

Production of Genetically Modified Foods –

- ✓ This is one of the widely debated application of rDNA.
- ✓ Genes can be derived from plants and even from animals to give plants characteristics that are beneficial to both producers and consumers, such as –
  - a) Delayed fruit ripening for longer shelf life.
  - b) Resistance to insects and plant varieties.
  - c) Enhanced flavour and nutritional content.
  - d) Edible vaccines to prevent the spread of diseases.



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